

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 325 — Past Questions & Materials
by Renance

Course: MEE 325

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — MEE 325

Level 300 • Second Semester • PastQuestion

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OF ENGINEERING AND TECHNOLOGY

PARINE CORROSION ENGINEERING (MAR 325)

RTMENT: MARINE ENGINEERING

SU ATTEMPT QUESTION 4, AND ANY OTHER 3

SEMESTER: SECOND SEMESTER

h responsible for the general,

k7 (2 Marks

ly susceptible to this form of attack

Sign of pitting corrosion. What will be the outcome

ow in a service environment subject to cyclical stresses? 2 Marks

es that could be implemented during a vessel's operation to prevent the
of pitting corrosion on the hull. (2 Marks)

process that occurs on an exposed hull surface in sea

-organisms are primarily responsible for its formation? (2 Marks)

'a component made of brass on a vessel undergoes selective corrosion?
the final state of the material. (2 Marks)

slatform with an area of 600cm² is exposed to air near the ocean. After

to experience a weight loss of 285g due to corrosion. To what rate of

yr does this correspond? (5 Marks)

these pits are left

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Viarks |

water. What is this

responsible for a highly branched cracks originating

@ type of corrosion cracking

tee] pipe. At what stress level can this occur? (2 Marks)

gasket is a classic example of crevice corrosion. Describe the specific

the crevice that initiates this attack. What role do chlorides play? (2

ves

Question 3 (15 Marks)

in a major metallurgical or affecting corrosion

a) Diagram the fundamental shape of a weld and structure does this property depend?

te. On what four specific aspects of
 @ Marks) ! | | the exchange current density of a
 in and, consequent, on 'ts errors!! is eat 'eatneat is used without subsequent
 Se aay hed. What will © auernal stresses, and the resulting corrosion rate?
 ae ifically regarding Ws 1 |
 a ta aeour between the ferrite and cementite
 cons a ucts the anode and which 2s the cathode, andi" 5
 la ie se bity. (2. 5 Marks) ee
 erwen significant deformation. ve MES
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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA
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COLLEGE OF ENGINEERING AND TECHNOLOGY

COURSE TITLE: MARINE CORROSION ENGINEERING (MAR 325)

DEPARTMENT: MARINE ENGINEERING

INSTRUCTIONS: ATTEMPT QUESTION 4, AND ANY OTHER 3

TIME: 2HRS SESSION: 2024/2025 SEMESTER: SECOND SEMESTER

Question 1 (15 Marks)

Describe the type of corrosion responsible for the general, even reduction in plate thickness. Why are

metals like steel particularly susceptible to this form of attack? (2 Marks)

b) Deep isolated pits are a sign of pitting corrosion. What will be the outcome if these pits are left untreated and continue to grow in a service environment subject to cyclical stresses? (2 Marks)

©) Discuss two specific measures that could be implemented during a vessel's operation to prevent the initiation and progression of pitting corrosion on the hull. (2 Marks)

d) Describe the initial biological process that occurs on an exposed hull surface in seawater. What is this

layer called, and which micro-organisms are primarily responsible for its formation? (2 Marks)

©) What will be the outcome if a component made of brass on a vessel undergoes selective corrosion?

Explain the process that leads to the final state of the material. (2 Marks)

f) A thick steel sheet of an offshore platform with an area of 600cm² is exposed to air near the ocean. After

a one-year period, it was found to experience a weight loss of 285g due to corrosion. To what rate of

corrosion, in both mpy and mm/yr does this correspond? (5 Marks) _ ee

a)

Question 2 (15 Marks)

2) Identify and explain the type of corrosion cracking responsible for a highly branched cracks originating

from pits in the stainless steel pipe. At what stress level can this occur? (2 Marks)

b) An attack under a PTFE gasket is a classic example of crevice corrosion. Describe the specific environmental condition inside the crevice that initiates this attack. What role do chlorides play? (2 Marks)

c) Discuss the mechanism of the corrosion found in the copper-nickel pipe at the bend. What is the specific

name for this type of damage, and how does the particulate matter in the fluid contribute to it? (2 Marks)

d) A general formula for corrosion can be expressed as Metal + Environment $\rightarrow$ Metal Compound + Free

Energy. Explain what the Environment contributes to the reaction for a metal like iron (Fe) or zinc (Zn)

in a marine setting. (2 Marks)

e) Describe one fundamental limitation in using protective coatings alone to prevent the mechanical failure

of a structure (2 Marks)

f) A large offshore platform with a surface area of 35000m² requires cathodic protection. The platform's

steel material requires a current density of 0.12mA/m², and the coating breakdown factor is estimated

to be 0.2. Zinc anodes with dimensions 0.5m by 0.16m will be used for protection. Assuming the anode

resistivity to the environment is 25 ohms.cm, anode electrochemical potential is -1.72V and the platform's hull potential is -0.73V, Determine the number of anodes needed for adequate protection. (5

Marks)

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2 >

Question 3 (15 Marks)

g corrosion

Give the fundamental material property that is the major metallurgical factor affecting, depend?

On what four specific aspects of the metal's composition and structure does this property effect that low Chromium and Nickel content will have on the exchange current density

Ola

Steel and consequently, on its corrosion rate in seawater (2.5 Marks)

c) A shaft has been quenched. What will be the outcome if this heat treatment is used without subsequent

- specifically regarding the phase stability, internal stresses, and the resulting corrosion rate?

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fa shaft underwent significant deformation. Describe why this
stz nt fo corrosion compared to the rest of the shafi, citing the two

like grain boundaries and second-

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iting corrosion resistance. Why

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA

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